

Small tunable X-ray sources may have large impact

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Imaging in the water window has traditionally relied on large synchrotron radiation sources. Now, a tabletop tunable X-ray source which generates water-window X-ray photons from a low energy electron beam interacting with a van der Waals crystal has been demonstrated.

Free-electron-based sources of electromagnetic radiation of various kinds have long attracted interest, primarily owing to their broad and continuous tunability, which is caused by the dependence of the radiation wavelength on the electron's kinetic energy. This is particularly true in the short-wavelength region of the spectrum, for example X-ray, where the availability and tunability of other radiation sources is very limited. The soft X-ray water window, the wavelength range (from around 4.4 nm to 2.2 nm) in which water is transparent but carbon is not, is a region rich with potential biological applications, for example imaging. Traditionally, synchrotrons have been used to generate radiation in this wavelength range, but these have the drawback of being extremely large in size, primarily owing to the accelerating structure required to produce high energy (MeV–GeV) beams of electrons.

By utilizing electron beams with energies of only a few keV, Nikhil Pramanik and co-workers at the Nanyang Technological University in Singapore have reported an alternative method of generating water-window X-ray photons with energies of several hundred eV that is much smaller in size than a synchrotron¹. In their system, shown schematically in Fig. 1, a beam of electrons is incident on a van der Waals (vdW) material, consisting of a series of 2D layers of atoms where the bonds between each layer are much weaker than the bonds between the atoms within each layer. X-ray photons are produced via two processes: parametric X-ray radiation (PXR) and coherent bremsstrahlung (CB). PXR occurs because of the layers of the vdW material, which behave as a grating and thus diffract the evanescent field of the incident electrons into propagating photons. CB emission occurs because of the interactions between the electrons and the spatially periodic lattice of the vdW material. Both processes result in photons with the same wavelength dependence on electron energy, tilt angle between the crystal zone axis and the electron beam propagation direction, and interlayer distance. The X-ray radiation produced is therefore continuously tunable via variation of the electron beam energy and tilt angle. Additional tunability is provided by the material composition, which determines the interlayer distance.

Pramanik and colleagues measured X-ray photon fluxes of 100–1,000 photons s^{−1} generated by a few-keV electron beam with a current of up to 50 nA. This photon flux is significantly lower than that required for most practical imaging and spectroscopy applications. However, Pramanik and colleagues also demonstrated that the X-ray

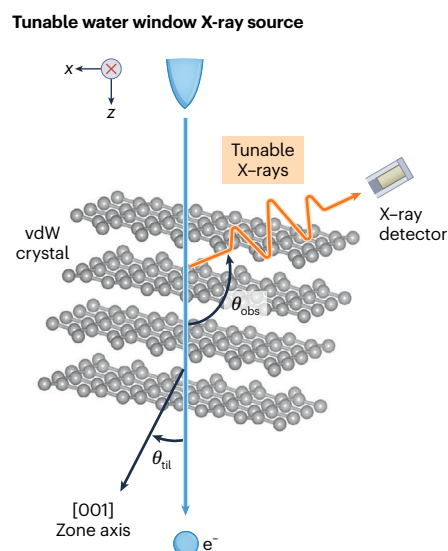


Fig. 1 | Tunable X-ray generation by an electron beam passing through a van der Waals material. A low energy (<keV) electron beam propagates through a van der Waals material consisting of multiple 2D layers, producing X-ray photons. Adapted from ref. 1, Springer Nature Ltd.

photon flux scales linearly with electron beam current and sub-linearly with the crystal thickness. Consequently, they predicted that much higher photon fluxes at values required for applications ($\sim 10^8$ photons s^{−1}) could be produced by electron beams with currents of around 10 mA. To support this prediction, Pramanik and colleagues carried out simulations of the electron–crystal interaction, which involved Monte Carlo simulations of electron scattering combined with electrodynamics simulations. As well as demonstrating good agreement with the results of their experiments, the simulations predicted that, by significantly increasing the electron beam current to around 10 mA, generation of photon fluxes around 10^8 photons s^{−1} would be possible without significant inhibition by either charging or heating of the crystal produced by the higher-current electron beam.

The results represent the intersection between two rapidly advancing fields: free-electron nanophotonics and van der Waals materials. Recently, free-electron nanophotonics has seen significant progress in developing tunable radiation sources based on relatively low energy electrons (relative to those used in traditional synchrotrons and free-electron lasers) with various nanophotonic structures (crystals, gratings, etc.)². Van der Waals materials have been the subject of intense interest in recent years, owing to their many unusual properties which arise from their 2D layered structure, for example extreme electro-magnetic confinement³ and linear energy-momentum dispersion⁴. Their potential as the basis of compact, resonant tunable radiation

sources has been recognized previously^{5–9} by exploiting the large mean free paths and interlayer atomic spacings unique to van der Waals crystals. The study by Pramanik and colleagues describes the first generation of X-rays from a vdW crystal in the technologically significant water-window spectral region from electron beams with energies of only a few keV.

If the X-ray flux scaling with electron beam current reported by Pramanik and colleagues can be sustained to the levels predicted, then the implications should be significant for biological imaging and spectroscopic applications in the water window region of the spectrum, and for other applications in this spectral region where the enormous photon fluxes generated by synchrotrons are not essential. More generally, recent advances in free-electron nanophotonics promise the realization of new regimes of electron–light interactions, involving, for example, quantum recoil of the electrons¹⁰ or topological effects¹¹. Although nanophotonic free-electron radiation sources have so far been based on spontaneous emission from independently emitting electrons, an important goal is the realization of nanophotonic free-electron lasers, which will be able to produce coherent stimulated emission via collective emission from many electrons.

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Competing interests

The author declares no competing interests.